

Azure Data Factory – Data Flows & Conditional Logic

Step 1: Verify the Resource Group

- Opened resource group DE_1, containing four resources: datafactoryde2612 (Data Factory V2), sqlserver1473569 (SQL database), sqlserver75893 (SQL server) and storage167562632 (Storage account).

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Microsoft.DataFactory-20260824114933 | Overview

DE_1 Resource group

Export resource groups using Bicep or Terraform | Help me generate Terraform for this resource group configuration. | How to manage changes with deployment tools?

Search

+ Create | Manage view | Delete resource group | Refresh | Export to CSV | Open query | Assign tags | Move | Delete | Group by none

Overview

Activity log | Access control (IAM) | Tags | Resource visualizer | Events | Settings | Cost Management | Monitoring | Automation | Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
datafactoryde2612	Data factory (V2)	Central India
sqlserver1473569 (sqlserver75893/sqlserver1473569)	SQL database	India South Central
sqlserver75893	SQL server	India South Central
storage167562632	Storage account	India South Central

Showing 1 - 4 of 4. Display count: auto

Give feedback

Step 2: Create and Populate the Employee Table

- In the SQL query editor, ran a script to create the EmployeeData table and insert sample records, then queried it back — 30 rows returned across columns like Department, JobTitle, Salary and Experience.

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Home > sqlserver1473569 (sqlserver75893/sqlserver1473569) | Query editor (preview)

sqlserver1473569 (sqlserver75893/sqlserver1473569) | Query editor (preview)

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Explorer

Search

sqlserver1473569

- dbo
 - Tables
 - EmployeeData
 - Views
 - Stored Procedures
 - Functions
- Queries
 - SQL query 1

SQL query 1

```
1 CREATE TABLE EmployeeData (EmpID INT PRIMARY KEY,FirstName VARCHAR(50),LastName VARCHAR(50),Gender VARCHAR(10),Age INT,Department VARCHAR(50),JobTitle VARCHAR(50),Salary INT,Experience INT);
2
3 INSERT INTO EmployeeData VALUES('Rahul', 'Sharma', 'Male', 28, 'IT', 'Software Engineer', 55000, 3, 'Mumbai', 101, 1);
4 select * from EmployeeData;
```

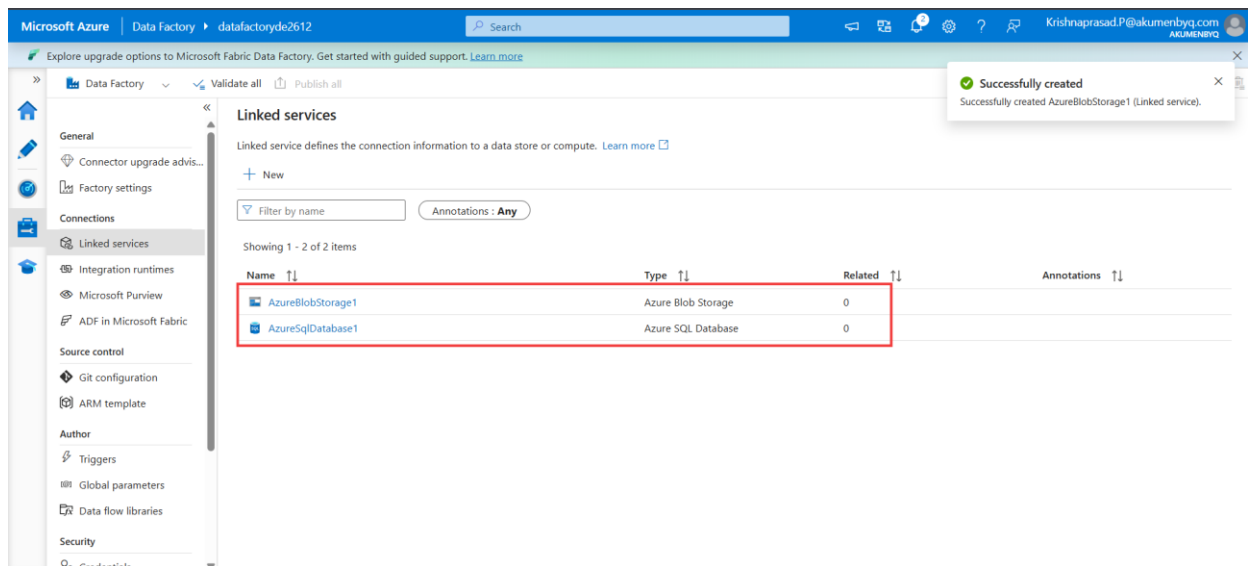
Messages Results

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle	Salary	Experience
1	Rahul	Sharma	Male	28	IT	Software Engi...	55000	3
2	Priya	Mehta	Female	32	HR	HR Manager	65000	7
3	Amit	Verma	Male	45	Finance	Accountant	48000	15
4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4
5	Karan	Singh	Male	35	Sales	Sales Executive	40000	8

Succeeded (0 sec: 180 ms) Columns: 12 Rows: 30

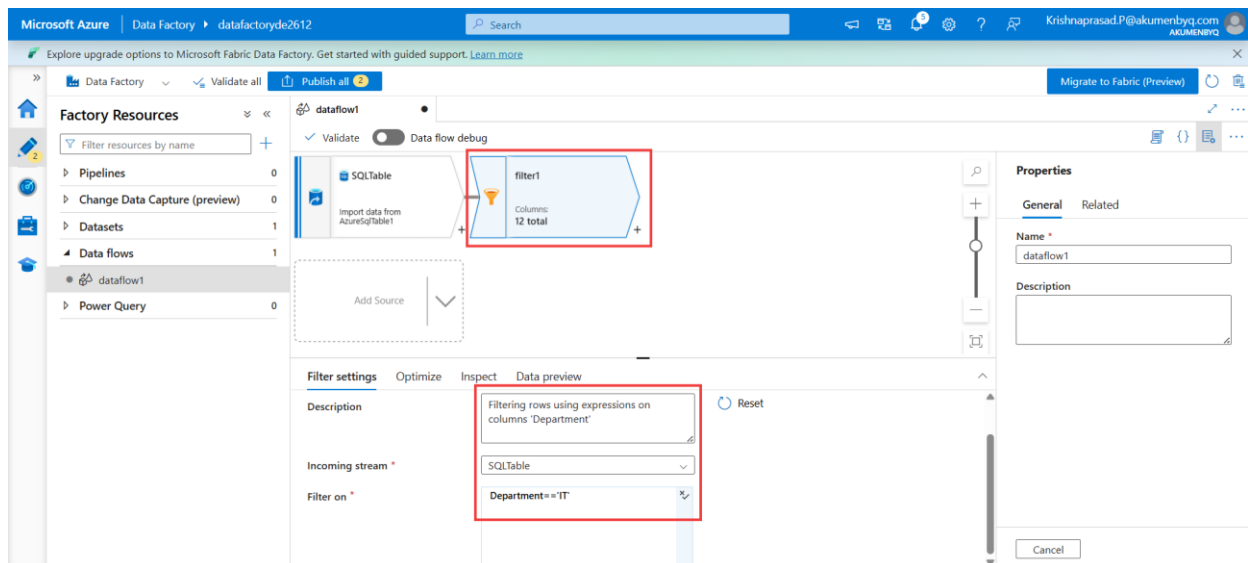
Step 3: Create the Linked Services

- In Data Factory, created two linked services — AzureBlobStorage1 (Azure Blob Storage) and AzureSqlDatabase1 (Azure SQL Database) — confirming successful creation.



Step 4: Filter Employees by Department

- Built a data flow (dataflow1) with SQLTable as the source, then added a Filter transformation (filter1) with the expression `Department == 'IT'` to keep only IT employees.



- Previewed the filtered data — 9 rows matched, including Rahul, Sneha and Rohit.

Microsoft Azure | Data Factory | datafactoryde2612

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
 - AzureSqlTable1
- Data flows 1
 - dataflow1
- Power Query 0

dataflow1

Validate Data flow debug Debug Settings

SQLTable Import data from AzureSqlTable1

filter1 Columns: 12 total

Filter settings Optimize Inspect Data preview

Number of rows: INSERT 9 UPDATE 0 DELETE 0 UPSERT 0

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle
1	Rahul	Sharma	Male	28	IT	Software Engineer
4	Sneha	Patil	Female	29	IT	Data Analyst
7	Rohit	Nair	Male	41	IT	Team Lead

Properties

General Related

Name * dataflow1

Description

Cancel

Step 5: Sort the Filtered Results

- Added a Sort transformation (sort1) after the filter, sorting by EmpID in ascending order.

Microsoft Azure | Data Factory | datafactoryde2612

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Factory Resources

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
 - AzureSqlTable1
- Data flows 1
 - dataflow1
- Power Query 0

dataflow1

Validate Data flow debug Debug Settings

SQLTable Import data from AzureSqlTable1

filter1 Filtering rows using expressions on columns 'Department'

sort1 Columns: 12 total

Sort settings Optimize Inspect Data preview

Incoming stream * filter1

Options *
☐ Case insensitive
☐ Sort only within partition

Sort conditions *

filter1's column	Order
EmpID	Ascending

Nulls first ☒

Open expression builder

- Previewed the sorted output — rows now ordered by EmpID (1, 4, 7, ...).

Microsoft Azure | Data Factory | datafactoryde2612

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Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 1
 - AzureSqlTable1
- Data flows: 1
 - dataflow1
- Power Query: 0

dataflow1

Validate Data flow debug Debug Settings

SQLTable Import data from AzureSqlTable1

filter1 Filtering rows using expressions on columns 'Department'

sort1 Columns: 12 total

Sort settings Optimize Inspect Data preview

Number of rows: INSERT 9 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle	Salary	Experience
1	Rahul	Sharma	Male	28	IT	Software Engineer	55000	3
4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4
7	Rohit	Nair	Male	41	IT	Team Lead	72000	12

Step 6: Configure the Sink Dataset

- Created a sink dataset (DelimitedText1) using linked service AzureBlobStorage1, writing to the output container with the first row as header.

Microsoft Azure | Data Factory | datafactoryde2612

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Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 1
 - AzureSqlTable1
- Data flows: 1
 - dataflow1
- Power Query: 0

dataflow1

Validate Data flow debug Debug Settings

SQLTable Import data from AzureSqlTable1

filter1 Filtering rows using expressions on columns 'Department'

sort1 Columns: 12 total

Sort settings Optimize Inspect Data preview

Number of rows: INSERT 9 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle	Salary	Experience
1	Rahul	Sharma	Male	28	IT	Software Engineer	55000	3
4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4
7	Rohit	Nair	Male	41	IT	Team Lead	72000	12

Set properties

Name: DelimitedText1

Linked service: AzureBlobStorage1

File path: output / Directory / File name

First row as header: ☒

Import schema: ☒ From connection/store ☐ From sample file ☐ None

Advanced

Sink type: Dataset

Incoming stream: sort1

Dataset: Select...

Options: ☒ Allow schema drift ☐ Validate schema

OK Back Cancel

Step 7: Run the Data Flow

- Ran Debug on the pipeline wrapping dataflow1 — the Data flow activity succeeded.

The screenshot shows the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane lists 'Pipelines' with 'dataflow' and 'Data flows' with 'dataflow1'. The main area shows the 'dataflow1' pipeline with a 'Data flow' activity. Below the pipeline, the 'Pipeline run' section shows a successful run with ID 'bc31e0a8-115a-48ac-8c4f-c4169809325c'. The 'Output' tab is selected, showing a table with columns: Activity name, Activity status, Activity type, Run start, Duration, and Integration runtime. The table contains one row for 'dataflow1' with a status of 'Succeeded'.

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
dataflow1	Succeeded	Data flow	8/24/2026, 12:21:15 PM	1m 16s	debugpool-8Cores-General-68 (Central I

- Verified the output file in blob storage: the filtered and sorted IT employee records were written as a CSV.

The screenshot shows the Microsoft Azure Blob Storage interface. The left pane shows the 'output' container. The main area displays a CSV file named 'part-00000-6e039c43-5119-4c75-9634-1edc98880bcd-c000.csv'. The file is previewed, showing a table with columns: EmpID, FirstName, LastName, Gender, Age, Department, JobTitle, Salary, Experience, City, ManagerID, JoinDate. The table contains 11 rows of employee data.

EmpID	FirstName	LastName	Gender	Age	Department	JobTitle	Salary	Experience	City	ManagerID	JoinDate
1	Rahul	Sharma	Male	28	IT	Software Engineer	55000	3	Mumbai	101	2021-05-10
4	Sneha	Patil	Female	29	IT	Data Analyst	52000	4	Bangalore	101	2020-08-20
7	Rohit	Nair	Male	41	IT	Team Lead	72000	12	Delhi	101	2012-09-10
11	Suresh	Menon	Male	38	IT	System Admin	50000	10	Bangalore	101	2014-07-19
14	Kavya	Shah	Female	28	IT	Backend Developer	56000	4	Pune	101	2019-06-15
18	Ritika	Soni	Female	27	IT	Frontend Developer	54000	3	Bangalore	101	2021-03-09
21	Lokesh	Bhat	Male	39	IT	Cloud Engineer	68000	11	Hyderabad	101	2014-02-17
27	Rohan	Kulkarni	Male	31	IT	DevOps Engineer	61000	6	Bangalore	101	2017-05-29
30	Reema	Jain	Female	29	IT	QA Engineer	50000	5	Chennai	101	2019-12-19

Step 8: Explore the Department Details Table

- Queried the DepartmentDetails table, returning six departments with their names and locations (IT/Bangalore, Finance/Pune, Sales/Chennai, Marketing/Mumbai, HR/Delhi, HR/Hyderabad).

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sqlserver1473569 (sqlserver75893/sqlserver1473569) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Overview | Activity log | Tags | Diagnose and solve problems | Query editor (preview) | Mirror database in Fabric (preview) | Resource visualizer | Settings | Data management | Integrations | Power Platform | Security | Intelligent performance | Monitoring | Automation

Explorer

Search

sqlserver1473569

dbo

Tables

DepartmentDe

EmployeeData

Views

Stored Procedures

Functions

Queries

SQL query 1

SQL query 2

SQL query 1

Run | Save as view

15 (101, 'IT', 'Bangalore'),

16 (102, 'Finance', 'Pune'),

17 (103, 'Sales', 'Chennai'),

18 (104, 'Marketing', 'Mumbai'),

19 (105, 'HR', 'Delhi'),

20 (106, 'HR', 'Hyderabad');

21 SELECT * FROM DepartmentDetails;

22

Messages | Results | () |

Department_ID	DepartmentName	Location
101	IT	Bangalore
102	Finance	Pune
103	Sales	Chennai
104	Marketing	Mumbai
105	HR	Delhi
106	HR	Hyderabad

Succeeded (0 sec 132 ms) Columns: 3 Rows: 6

Step 9: Join Employees with Department Details

- Built a new pipeline (join) with a data flow joining source1 (employee data) and source2 (department data), using an Inner join on Department = DepartmentName.

Microsoft Azure | Data Factory | datafactoryde2612 | Search

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Validate all | Publish all | Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines

Join

Change Data Capture (preview)

Datasets

AzureSqlTable1

AzureSqlTable2

DelimitedText1

Data flows

dataflow1

Power Query

dataflow1

Validate | Data flow debug | Debug Settings

source1

Import data from AzureSqlTable1

source2

Import data from AzureSqlTable2

join

Columns: 15 total

sink1

Export data to DelimitedText1

Join settings | Optimize | Inspect | Data preview

Join type *

Full outer | Inner | Left outer | Right outer | Custom (cross)

Use fuzzy matching

Join conditions *

Left: source1's column

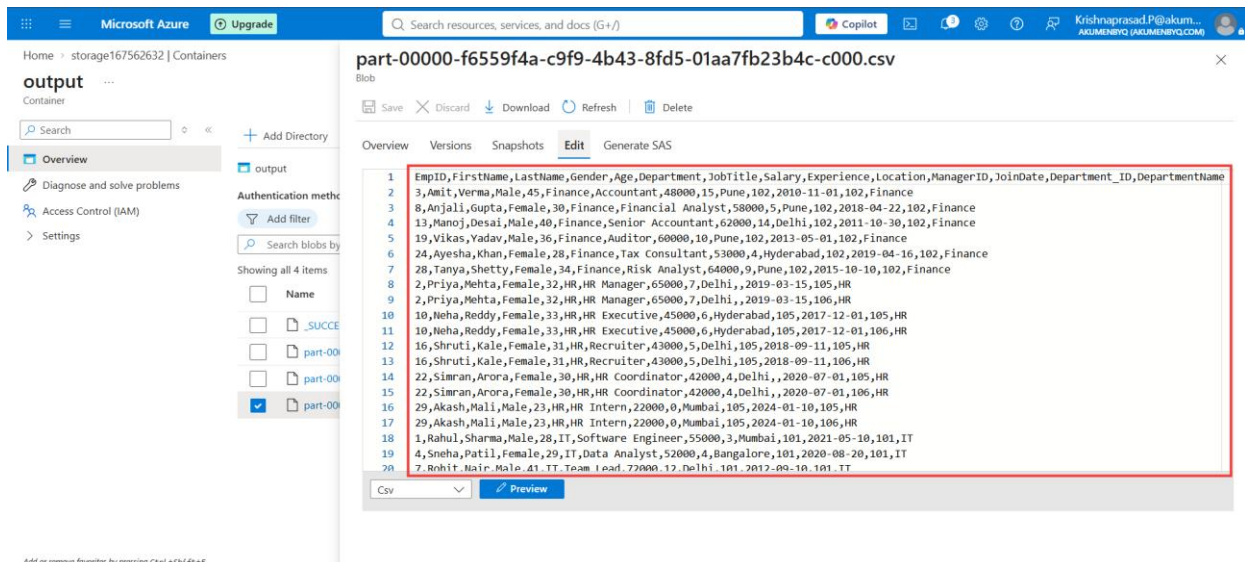
Right: source2's column

abc: Department

==

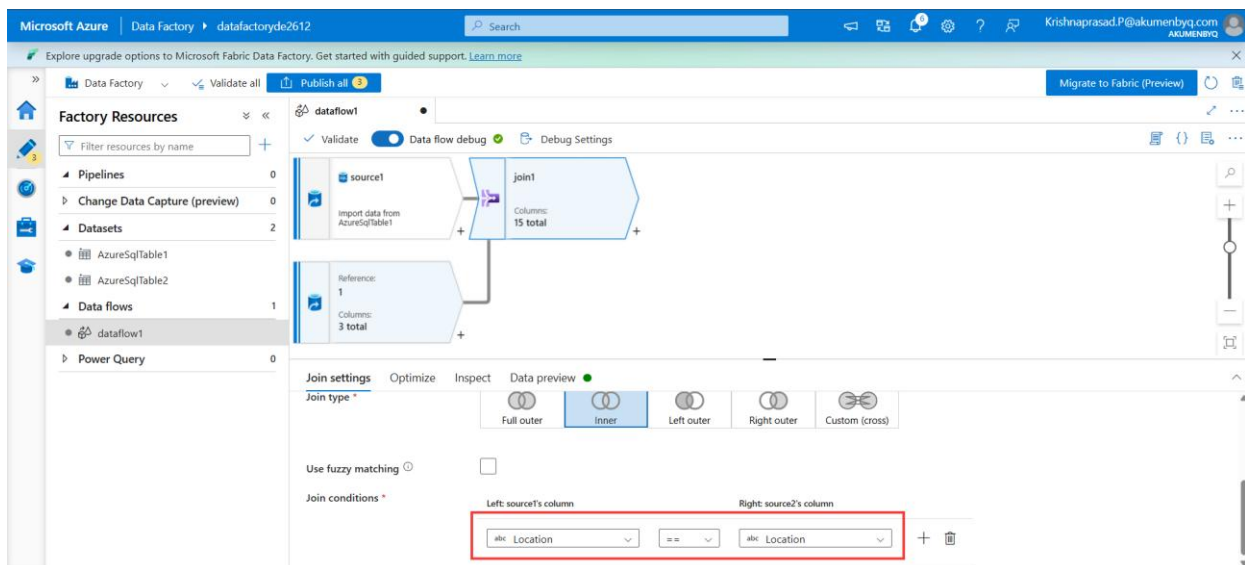
abc: DepartmentName

- Verified the joined output file in blob storage, combining employee records with their department name and location.



Step 10: Re-Join on Location

- Changed the join condition to match on Location = Location instead, to see employees and departments that share the same city.



- Previewed the results — rows where both sides' Location matched (Mumbai, Delhi, Pune, Bangalore).

Microsoft Azure | Data Factory | datafactoryde2612

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Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 2
 - AzureSqlTable1
 - AzureSqlTable2
- Data flows: 1
 - dataflow1
- Power Query: 0

dataflow1

Validate Data flow debug Debug Settings

Join settings Optimize Inspect Data preview

	Location	Location	ManagerID	JoinDate	Department_ID	DepartmentName
re Engineer	Mumbai	Mumbai	101	2021-05-10	104	Marketing
haeder	Delhi	Delhi	NULL	2019-03-15	105	HR
stant	Pune	Pune	102	2010-11-01	102	Finance
nalvst	Bangalore	Bangalore	101	2020-08-20	101	IT

Step 11: Configure the Join Sink

- Set the sink for the join data flow to the DelimitedText1 dataset, taking input from the join1 stream.

Microsoft Azure | Data Factory | datafactoryde2612

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Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 3
 - AzureSqlTable1
 - AzureSqlTable2
 - DelimitedText1
- Data flows: 1
 - dataflow1
- Power Query: 0

dataflow1

Validate Data flow debug Debug Settings

Sink Settings Errors Mapping Optimize Inspect Data preview

Incoming stream * join1

Sink type * Dataset Inline Cache

Dataset * DelimitedText1 Test connection Open New

Skip line count

Options ☒ Allow schema drift

Step 12: Run and Verify the Join Pipeline

- Ran Debug on the join pipeline — the Data flow activity succeeded.

The screenshot shows the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane lists 'Data flows' with 'dataflow1' selected. The main canvas displays a 'Data flow' activity named 'dataflow1'. Below the canvas, the 'Pipeline status' is 'Succeeded'. A table shows the run details:

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
dataflow1	Succeeded	Data flow	8/24/2026, 12:56:10 PM	1m 26s	debugpool-8Cores-General-68 (Central I

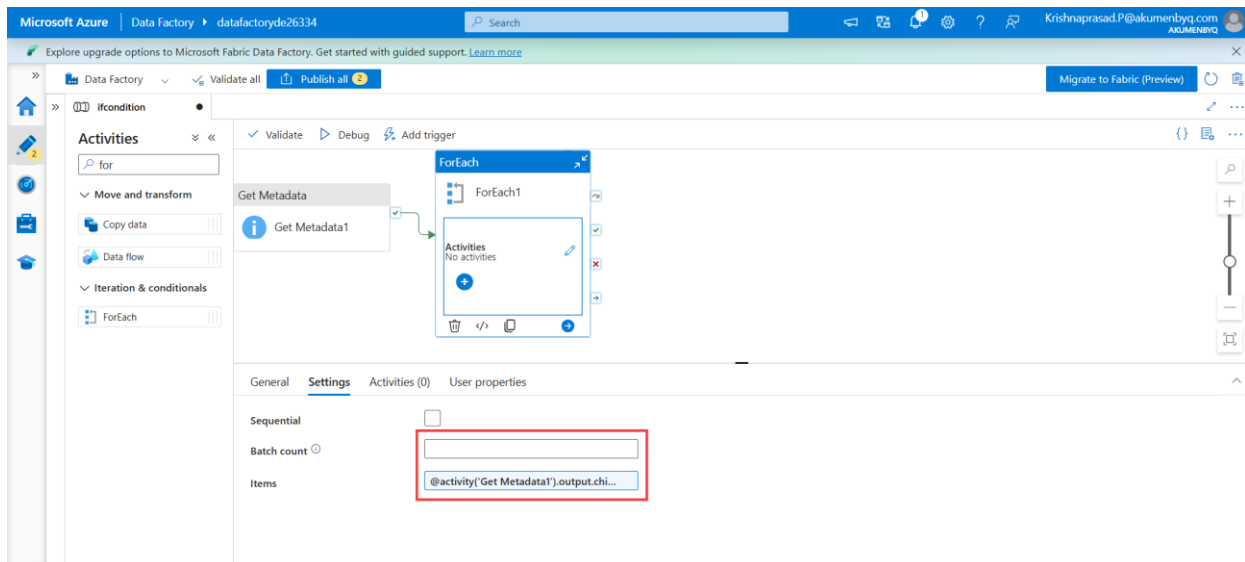
Step 13: Get Metadata for the Output Folder

- Created a new pipeline (ifcondition) with a Get Metadata activity pointed at the DelimitedText1 dataset to list the folder's contents.

The screenshot shows the 'Get Metadata' activity configuration in the Microsoft Azure Data Factory console. The activity is named 'Get Metadata1' and is configured with the following settings:

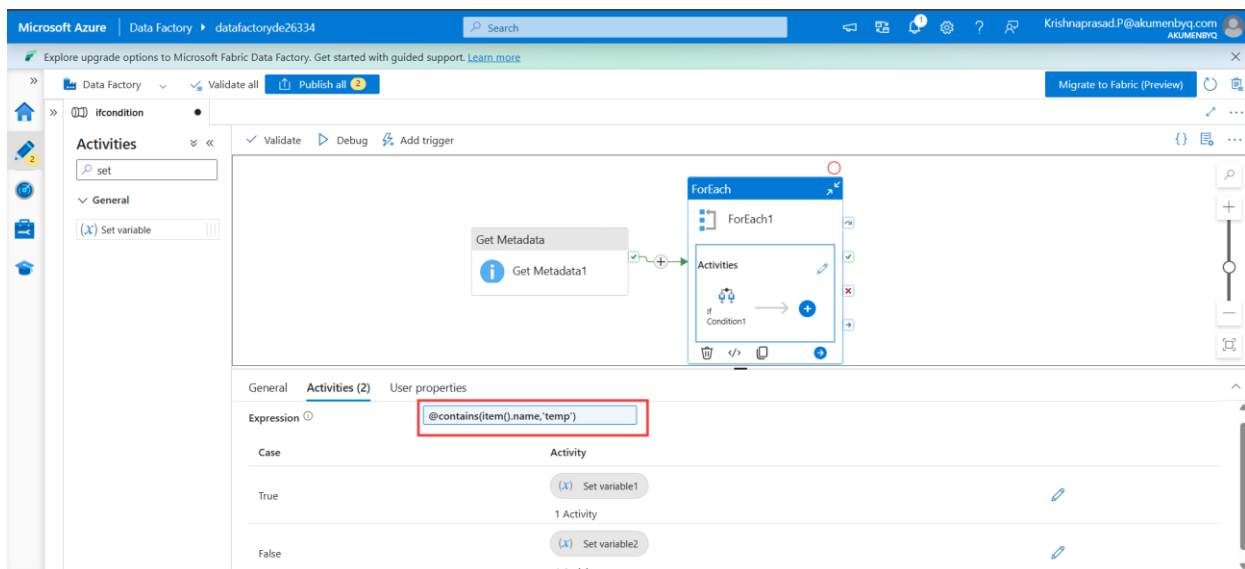
- Dataset:** DelimitedText1
- Field list:** + New
- Filter by last modified:** (checkbox checked)
- Start time (UTC):** (empty field)
- End time (UTC):** (empty field)
- Skip line count:** (empty field)

- Added a ForEach activity after Get Metadata, iterating over the child items returned: `@activity('Get Metadata1').output.childItems`.



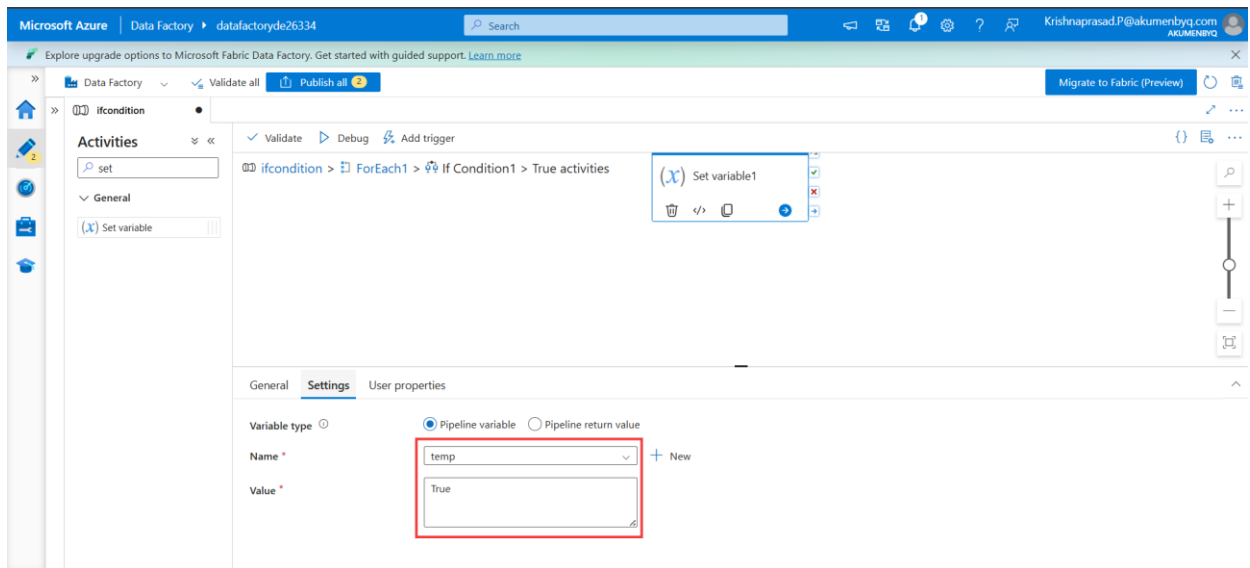
Step 14: Branch Logic with If Condition

- Inside the ForEach loop, added an If Condition activity that checks whether each file name contains "temp": `@contains(item().name,'temp')`.

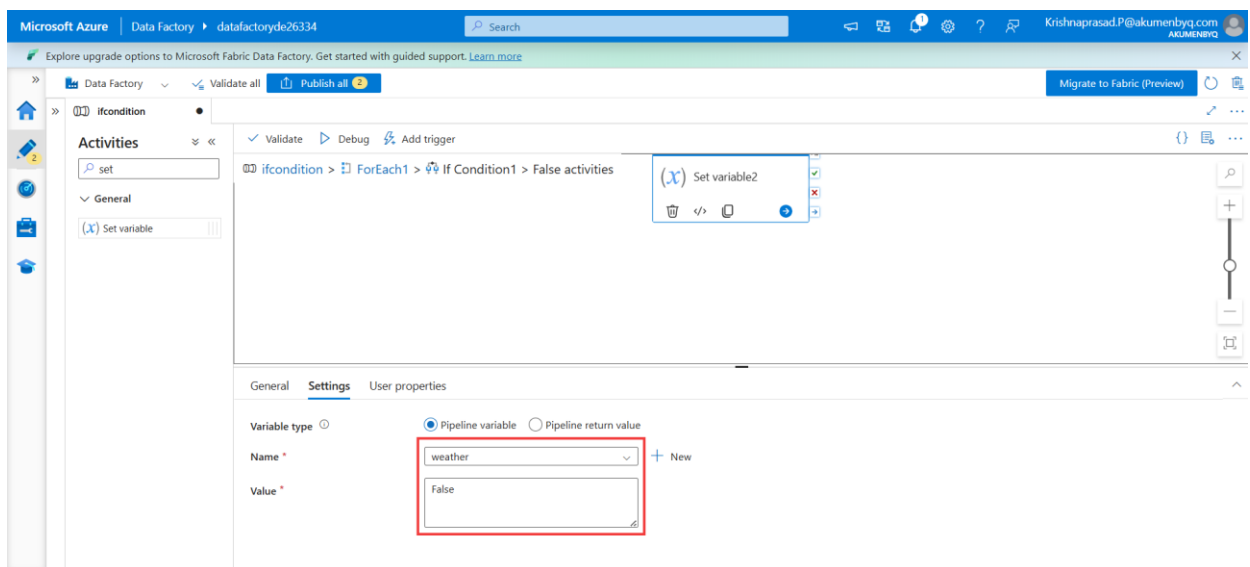


Step 15: Set Variables for Each Branch

- In the True branch, added a Set variable activity setting temp to "True".



- In the False branch, added a Set variable activity setting weather to "False".



Step 16: Inspect a Sample File

- Opened one of the files being iterated over — a weather.csv containing date, temperature, humidity and condition columns — confirming why its name didn't match the "temp" filter and fell into the False branch.

Microsoft Azure

Upgrade

part-00000-59535516-a679-4423-a58a-4bab5cf849d3-c000....

Blob

Save

Discard

Download

Refresh

Delete

Overview

Versions

Snapshots

Edit

Generate SAS

1	date,temperature_c,humidity_pct,condition,temp_F
2	2026-08-19,22,60,Sunny,71.6
3	2026-08-20,18,75,Cloudy,64.4
4	2026-08-21,25,68,Rainy,77.0
5	2026-08-22,19,55,Clear,66.2
6	2026-08-23,30,80,Humid,86.0
7	

Csv

Preview